Measurement Beyond Survey Responses, Part B

Social Media

Surv/SurvMeth 621

Fundamentals of Data Collection

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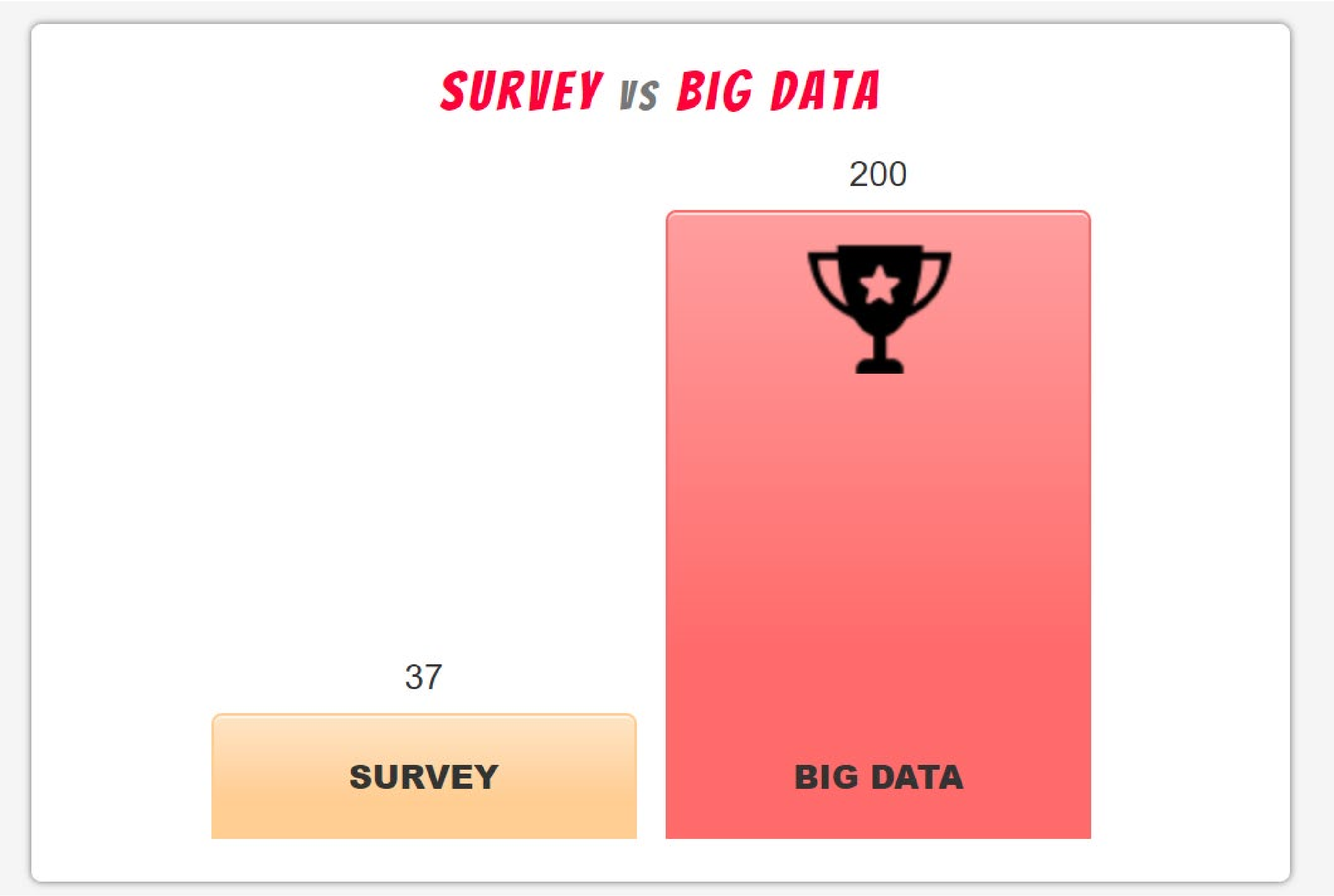
Self-reports vs. Found Data

- Most surveys ask *R*s questions and their answers become the data
- Self-reports can concern almost any topic of interest to researchers
  - whether *R*s have thought previously about the topic, if subjective
  - engaged in activity, if behavioral

Survey vs. Big Data: Google fight? (2019)



In Class Exercise

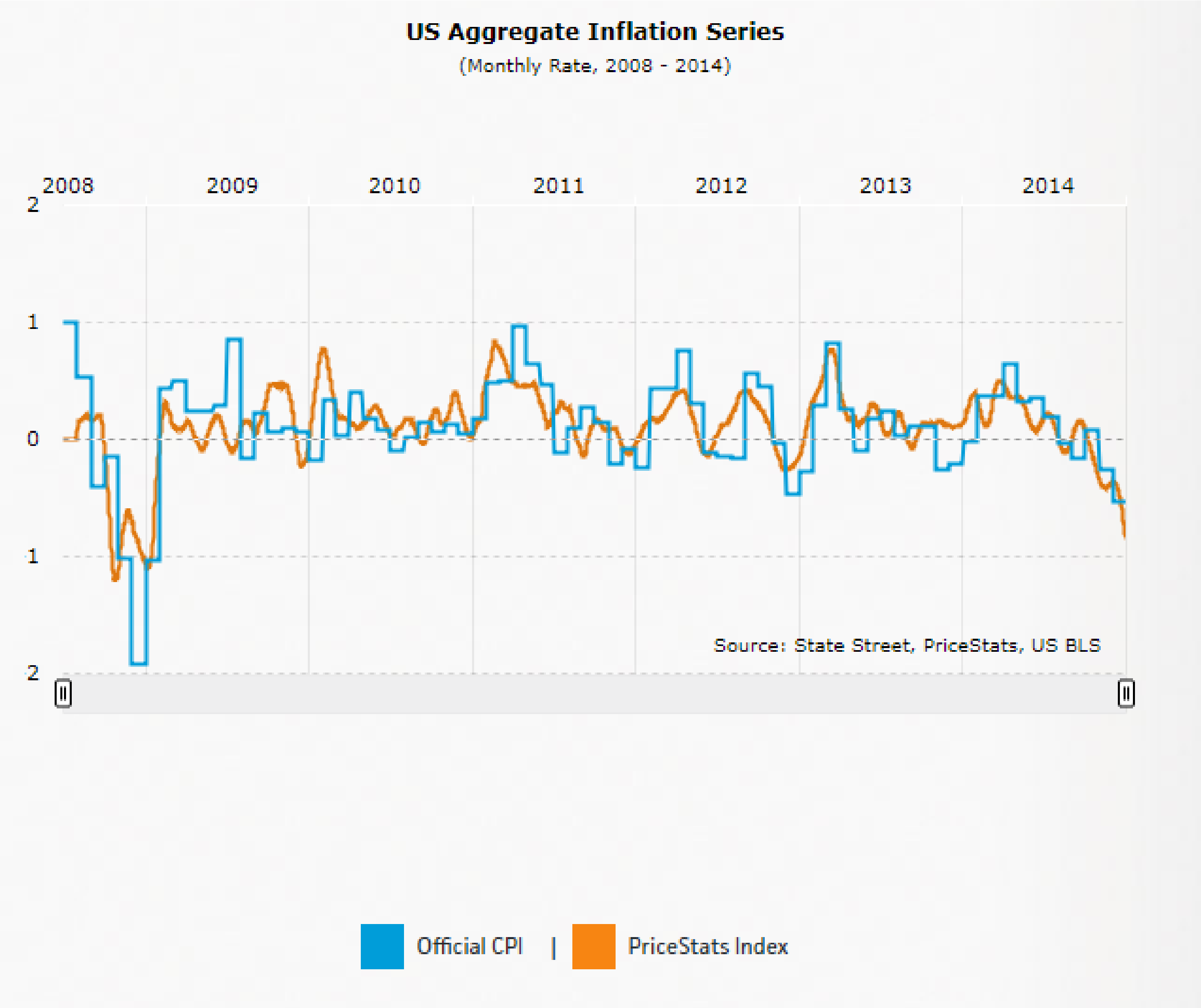
- Groups are assigned in Inclass Exercises
  - I will reassign zoom participants during the class
- Review the information on slides 5-7
- Compare (Note what is different) BLS's CPI Change and PriceStats' Inflation Estimates using Total Survey Error framework (Please think about Representation and Measurement in particular)
- Compare two series by goodness of fit measures such as timeliness, transparency, etc. and ethics in terms of data privacy and confidentiality

Survey vs. Big Data: BLS Consumer Price Index (CPI) vs. Pricestats Inflation Series

- BLS CPI is based on survey data (https://www.bls.gov/cpi/questions-and-answers.htm)
  - The Consumer Price Index (CPI) is a measure of the average change over time in the prices for a market basket of consumer goods and services paid by urban consumers .
  - 12,000 consumers/year keep diaries listing everything they bought during a 2-week period

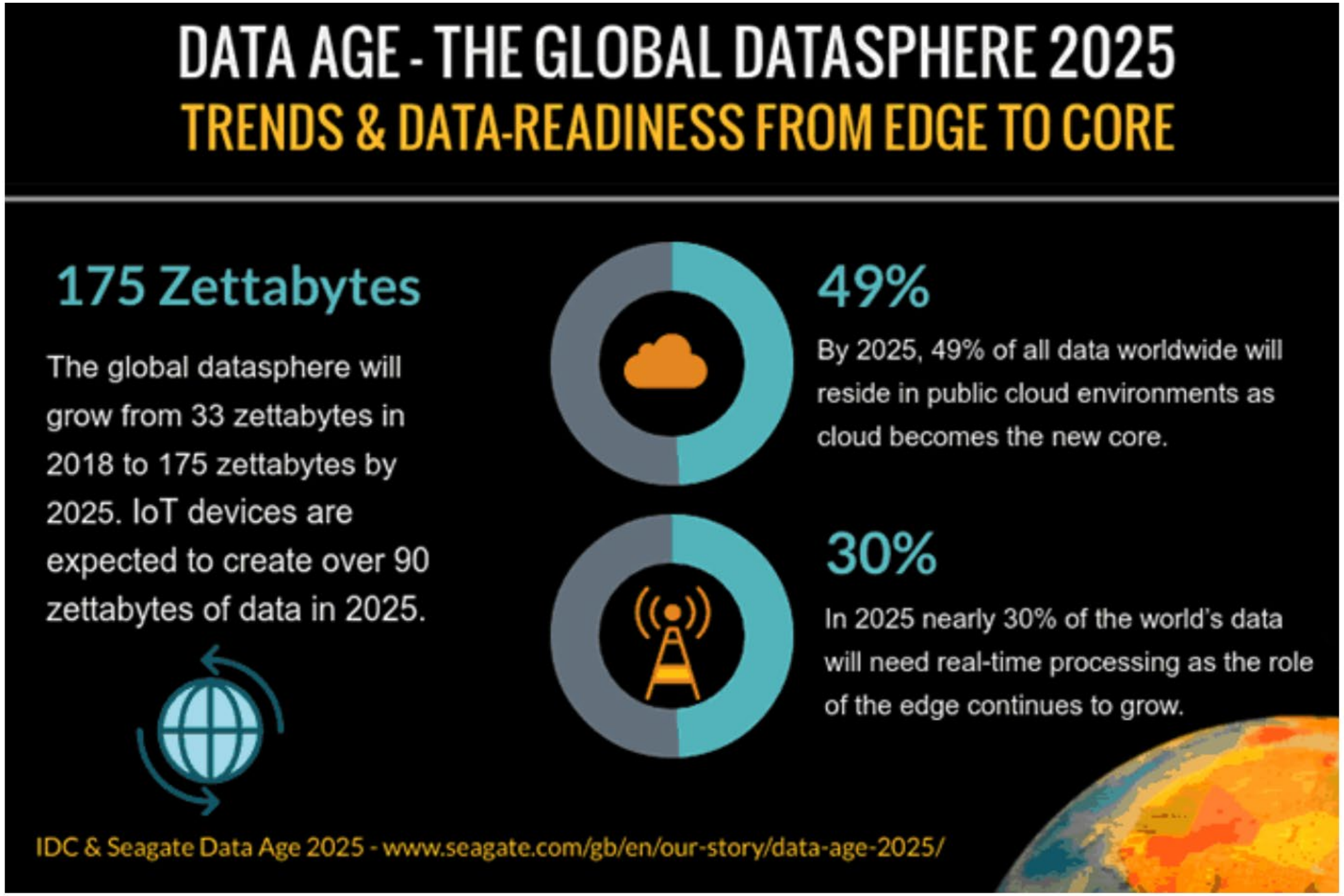
- Pricestats' Inflation Series are based on web data
  - Daily available data
  - Selected group of online retailers

Survey vs. Big Data: BLS CPI vs. Pricestats Inflation Series (https://www.pricestats.com/inflation-series?chart=1837)



Self-reports vs. Found Data (2)

- We live in a data rich world
  - Search strings, social media posts, step counts, streaming …
  - Massive data sets
- But most data are *organic,* i.e., created as biproduct of everyday life
- Not *designed* for addressing research questions
  - Groves (2011)
- To use these "found" data, they must be repurposed



https://www.i-scoop.eu/big-data-action-value-context/

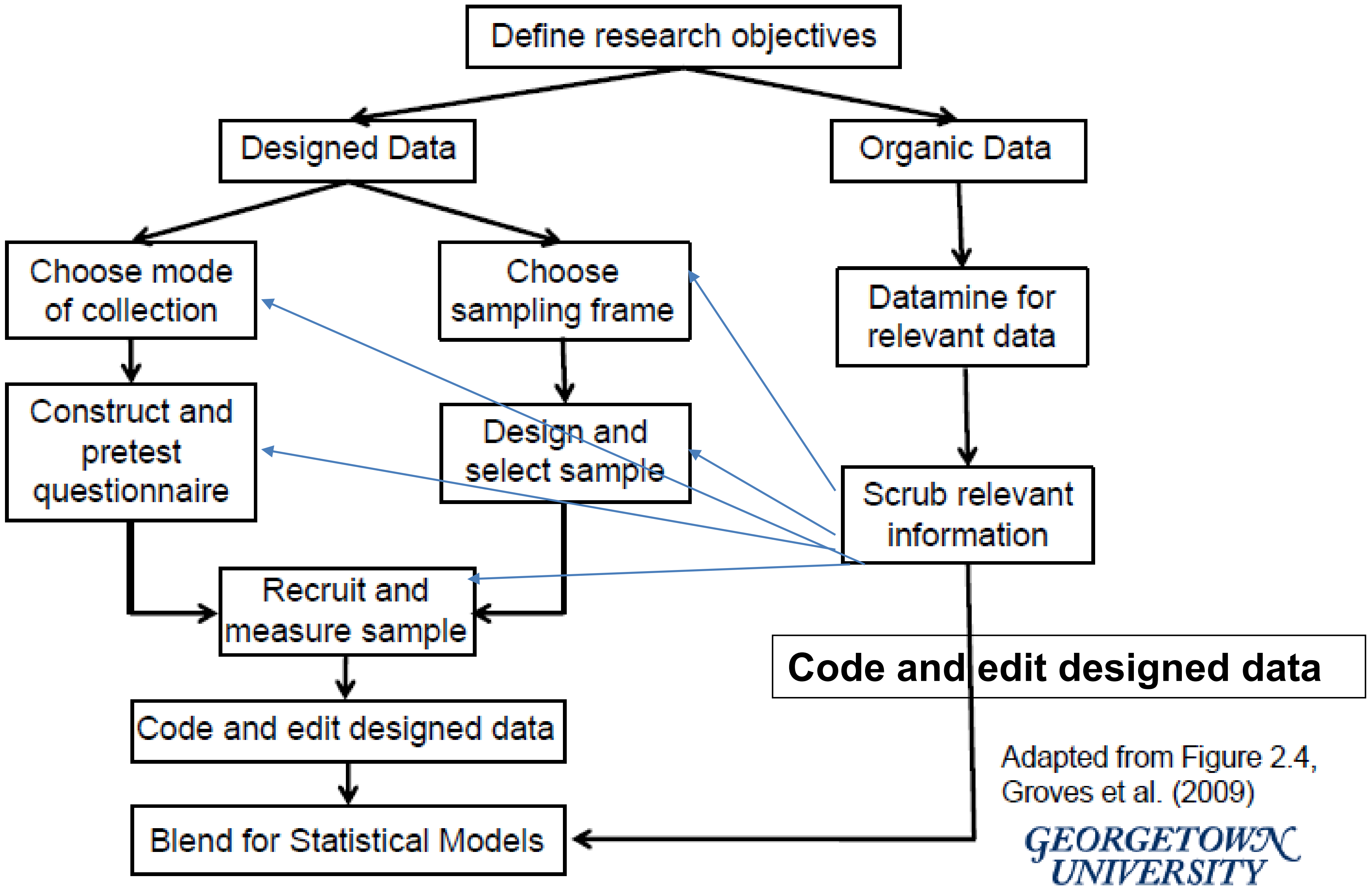
The Survey Research in the Big Data World

- Big Data (or organic data)
  - https://www.i-scoop.eu/big-data-action-value-context/
- Survey Data
  - Designed
  - Total Survey Error framework

Two Dominant Schools of Thought

- Big data as an alternative to survey data
  - There is a need for a quality framework for organic data (Groves, 2011)
  - Some researchers already responded to this call:
  - Hsieh, Y. P., Murphy, J. (2017). Total Twitter error: Decomposing public opinion measurement on Twitter from a total survey error perspective. In Biemer, P. P., de Leeuw, E., Eckman, S., Edwards, B., Kreuter, F., Lyberg, B. T., West, L. E. (Eds.), Total survey error in practice: Improving quality in the era of big data (pp. 23–46). Hoboken, NJ: John Wiley & Sons.
- Big data as supplement to survey data
  - *The combination of designed data with organic data is the ticket to the future.* Director Robert Groves, May 31, 2011.
  - This type of thinking is not foreign to survey research

Blending Survey and Organic Data (Groves 2011)



Adapted from Figure 2.4, Groves et al. (2009)

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Some Kinds of Found/Organic/Big Data

- Search strings
- Online prices
- Administrative records
- Social media ⟵ Focus on measurement issues
- Geolocation (of mobile phones)
- Nighttime luminosity

Can Social Media Content Be Incorporated Into Social Research?

- AAPOR Task Force on Big Data (2014): to obtain *qualitative* insights and monitor trends, *not* to obtain accurate, *quantitative* estimates
- Fit for use?
  - method/type of data must be able to accurately answer research question
- Probably okay to:
  - Monitor trends, e.g., attitudes toward economy or a company's brand image
  - Compare quantities, e.g., relative amount of interest in a sporting event in different locations
  - Develop qualitative insights as in traditional, in-person focus groups
- Probably not okay to:
  - Estimate prevalence, e.g., of childhood obesity in U.S.
  - Attribute causality to relationship between one variable and another, e.g., whether a public health campaign reduced prevalence of a disease

Social Media for Social Measurement: Opportunities

- Cost: **Probably** lower than conducting most surveys
- Timely and continuous; volume of posts sensitive to public events
- Users are motivated to post content without prompting by researchers
- No burden on users beyond what they voluntarily invest

Social Media for Social Measurement: Obstacles to Adoption

- Researchers do not drive topics or timing of posts
  - Unlikely, even in massive corpus, there will be enough posts on all, or even most, topics of interest to conduct credible analysis
  - What leads to a post is usually unknown to researchers
- Little user-level info
  - Some can be extracted from content, e.g., age, gender (Barberá, 2016), political orientation (Cohen & Ruths, 2013)
  - Geo-tagging is available in some social media (for some users) but many users do not turn it on (Jensen, 2017)
  - Virtually no covariates (“case-rich, variable poor”) – typically counting volume or average sentiment of posts on a topic on a day
- Often requires substantial pre-processing, which is expensive

Social Media for Social Measurement: Obstacles to Adoption (2)

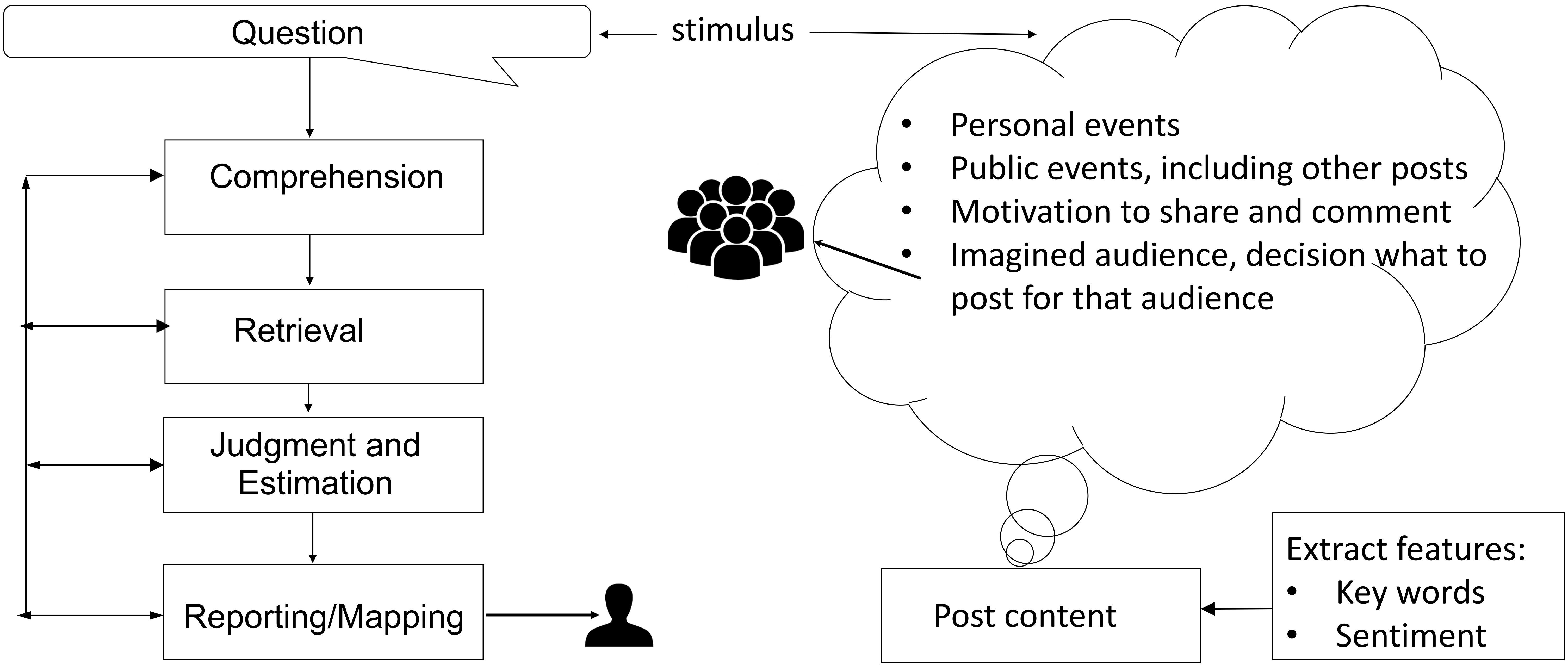
- Automated textual analysis has not, so far, been very nuanced
  - Complicated by use of pronouns, negation, metaphors, irony, slang
- Spurious correlations
  - Common in time series data (http://www.tylervigen.com/spurious-correlations).
  - e.g., per capita cheese consumption and number of people who died by becoming tangled in their bed sheets (r = .95)
- Despite the obstacles, much enthusiasm about promise of social media as one type of data for social research

Representation: Elephant in the Room

- Coverage issues may prevent generalization beyond platform
  - Massive corpus does not assure coverage of population (”Big is not all”)
  - Likely biases for demographic factors, e.g., age and behaviors
- But our focus, today, is on measurement

Social Media Content as Data

1. Extract content – usually text – from site
   – Scraping: collecting content from web sites, e.g., a table on a web page
   – API: created by social media site to provide public data sets of user content
   – Commercial Providers, e.g., Sprinklr, Hootsuite, Sysomos
2. Extract meaning from text (often filtered by content)
   – Frequency of posts, especially that contain particular keyword(s)
   – Sentiment (pos. vs. neg.): e.g., Lexicoder, OpinionFinder, Vader, Textblob
   – Lexical (types of words such as emotion): e.g., LIWC – often reduces to sentiment
   – Topic/semantic similarity to a target: e.g., Language modeling tools like Bert
   – Stance (opinion of posts): e.g., Bert-like tools and LLMs



Survey Response Process

Social Media Posting Process

Outline for Remainder of Class

- Imagined audiences and selective self-presentation
- Social media as standalone data
  - i.e., when survey data not available or social media better measures phenomena of interest
- Social media as supplement to survey data
- Social media as alternative to survey data

Origins of Social Media Content

- Social media users design their posts for others
  - Imagined audience
- When users post, they often want audience to like or at least notice them and their posts
- Selective self-presentation
  - Users post content that will promote these goals
  - Less likely to post content that will not promote goals

How Twitter Users Imagine Their Audience

- Marwick & boyd (2010) asked Twitter users who they imagined they were tweeting to
- Goal was to elicit diverse set of responses – not to quantify frequency of views
- Asked
    1. Their own followers
    2. Random sample of users in public timeline, some with large and some with small numbers of followers

Imagined Audiences (2)

- Some users claim to not consider audience:

  "I don't tweet to anybody; I just do it to do it."

  "when I tweet, I tweet honestly, I tweet passionately. Pure expression of my heart."

  "Honestly, I have no idea who reads them. Hopefully a very small group of very forgiving people!"

- Others do have an audience in mind:

  "I think of a room filled with friends when I tweet. I assume people like me that are reading my tweets."

Imagined Audiences (3)

- Some users are more strategic, posting content that they believe will engage imagined audience:
  > "U know I don't know who reads them, but when I tweet sumthin contrvsial or interesting I find a get a cple more followers."
- Some users self-censor:
  > "Won't Tweet anything too personal, TMI about self/others, dead horse areas from subjects like religion/politics/sports"

  > "i'm very conscious that twitter is public. i wouldn't tweet anything i didn't want my mother/employer/professor to see"

  > "I tweet about anything I would say in a lobby. Beyond that, each tweet is influenced by the tweets around that unique moment."

Imagined Audiences (4)

- Not only do users imagine their audiences based on little evidence, but they are inaccurate in estimating audience size
- Bernstein et al. (2013) surveyed active Facebook users and asked

  "How many people do you think usually see the content you share on Facebook?"
- or, for a specific post

  "How many people do you think saw it?"
- Users systematically underestimated size of audience compared to Facebook logs
  - Estimates based on guesses, likes and comments, number of friends, etc.

Imagined Audience (5)

- How might the way users think about their audience affect measurement?
- Does accuracy of posts depend on imagined audience?
  - Budak (2019) showed that tweets about Clinton and Trump – especially Clinton – in 2016 presidential election contained more fake news* as campaign progressed
  - Presumably, users posted more fake news as electorate became more polarized and their demand for extreme content increased
  - May reflect what users think, i.e., content may not have been knowingly fake, but may be intentionally misleading
    - hard to know on face of it

*based on classification of publishers by Alcott et al., (2019), National Bureau of Economic Research

Measuring Mood Across The Day (2)

- Golder and Macy (2009) analyzed affect in 509 million tweets from 84 countries from Feb 2008 - Jan 2010
  - Classified affect in all tweets as Positive (PA) or Negative (NA) and tallied for each day
  - Vastly increases temporal resolution (content posted around the clock); immediacy of posts reduces recall error
  - 2.4 million users is very large sample, but they acknowledge it is not guaranteed to represent the 84 national populations

Measuring Moods Across The Day (3)

PA peaks early morning and at midnight on weekdays; peak higher and later on weekends; NA lower in morning and independent of PA

PA and NA patterns across countries/regions resemble aggregate patterns (on left); differ statistically from each other (lots of power!)

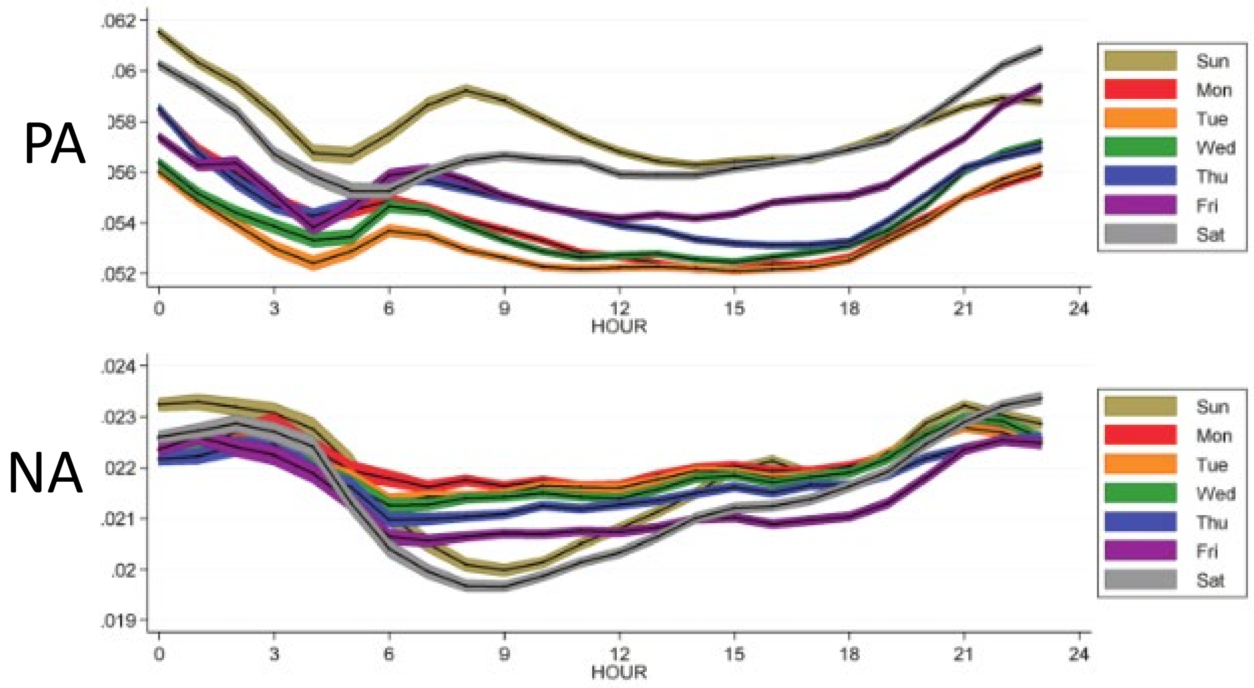


Fig. 1. Hourly changes in individual affect broken down by day of the week (top, PA; bottom, NA). Each series shows mean affect (black lines) and 95% confidence interval (colored regions).

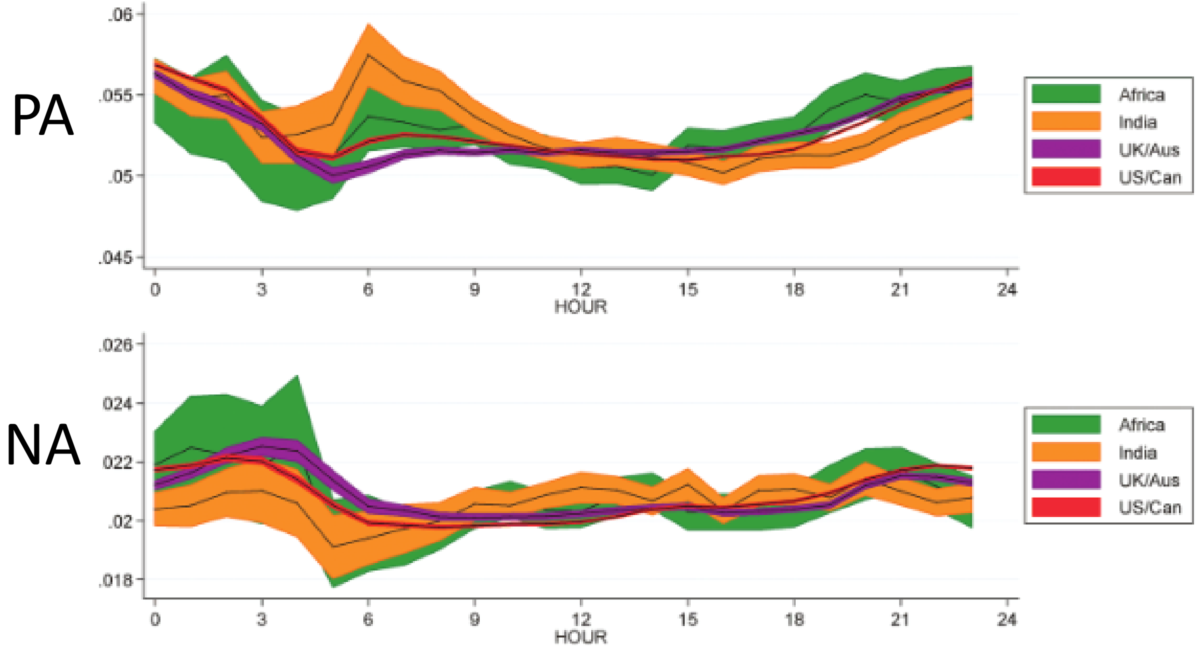


Fig. 2. Hourly changes in individual affect in four English-speaking regions. Each series shows mean affect (black lines) and 95% confidence interval (colored regions).

Golder and Macy (2009), *Science*

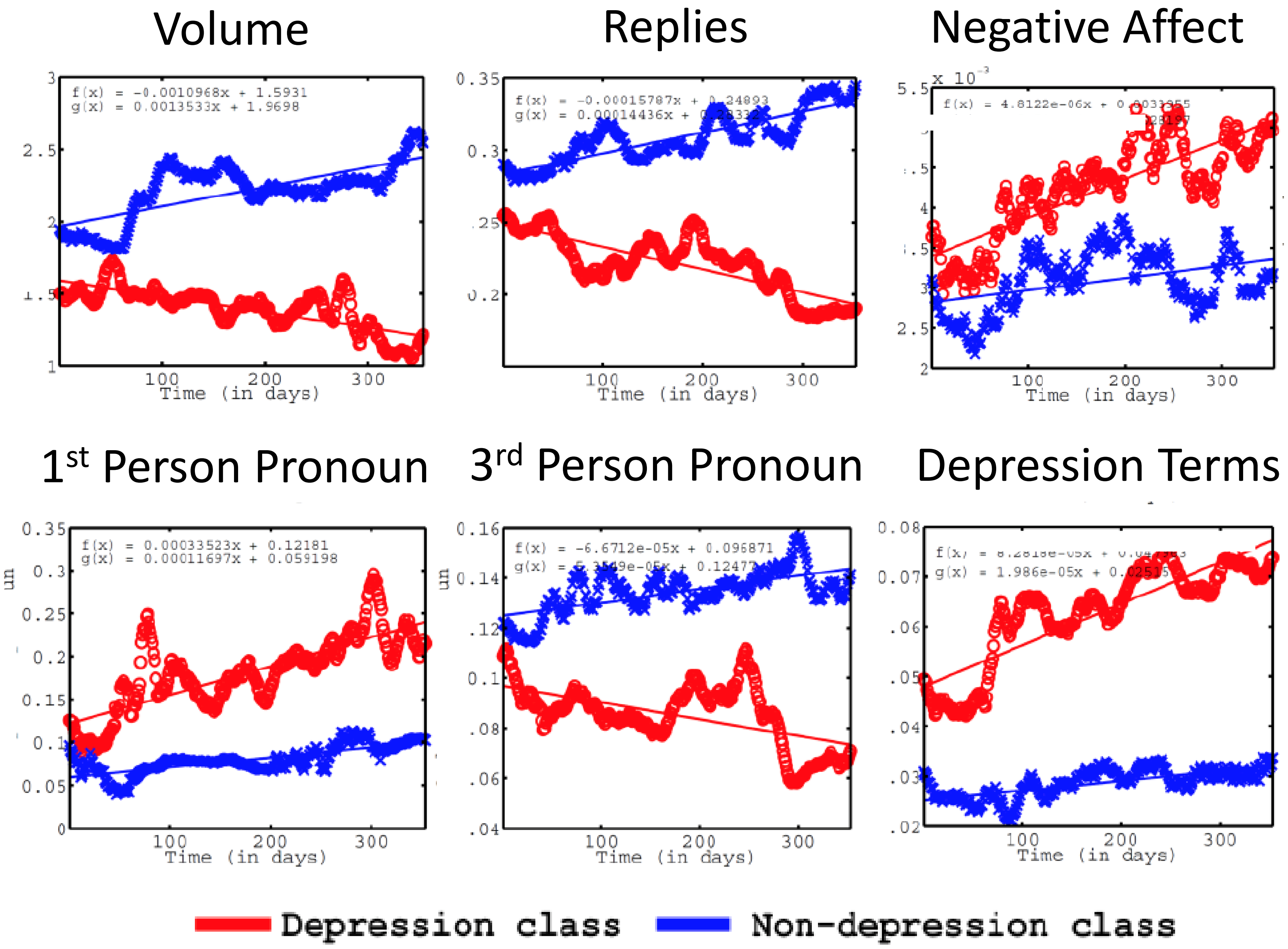
Social Media for Detecting Depression

- De Choudhury et al. (2013) compared 1 year of tweets posted by depressed and non-depressed users to develop tool for detecting depression before diagnosis
- Participants (MTurkers) completed a depression index and authorized access to their tweets

Social Media for Detecting Depression (2)

- Two classes of participants:
  - Depressed (n=171): scored in the moderate to high depression range <u>and</u> reported being diagnosed for Major Depressive Disorder (MDD
  - Non-depressed (n=305): Scored in the low/no depression range

Tweets by Depressed and Non-depressed Users



Source: De Choudhury et al. (2013)

Social Media for Detecting Depression (3)

- De Choudhury et al. developed a model to predict diagnosis of MDD based on these and other features of users' tweets
- Successfully classified users 70% of the time
- They propose that if scaled up, could be used for automated public health tracking, computing MDD-risk score for members of the public (at least those who use Twitter)

Social Media as Supplement to Survey Data

- Smith and Gustafson (2017) incorporated Wikipedia pageviews for candidates in ~100 US Senate races (2008- 2012) into projection models
  - Argue that candidate pageviews indicate liking a candidate and probably indicate increased likelihood of voting for the candidate
- Compared models that did not include and did include pageviews:
  - "Simple models" comprised of poll results (proportion vote share of Democratic candidate) and fundamentals (e.g., presidential approval, incumbency, economic indicators)
  - "Synthesized model" included variables in simple model plus (ln transformed) # candidate pageviews; tested in 2-week intervals from 28 to 2 weeks before election
- Incorporating pageviews significantly improved fit of model at all but one interval

Projections With and Without Pageviews

Table 2. Results of Synthesized Projection Model, Combining Fundamentals, Polls and Wikipedia Pageviews Projections

	Model 1				Model 2						
			Fund.	Polls					Fund.	Polls	Pageviews
	R^2	RMSE	b^*	b^*	R^2	ΔR^2	RMSE	ΔRMSE	b^*	b^*	b^*
28 Weeks	.757	6.55	.409***	.499***	.773	.016*	6.34	–0.22	.320**	.407***	.213*
26 Weeks	.777	6.28	.408***	.519***	.795	.018**	6.02	–0.25	.301**	.437***	.224**
24 Weeks	.787	6.13	.409***	.528***	.803	.016**	5.90	–0.24	.306***	.448***	.215**
22 Weeks	.789	6.10	.383***	.550***	.802	.013*	5.91	–0.20	.296***	.472***	.195*
20 Weeks	.806	5.85	.355***	.587***	.812	.006	5.77	–0.09	.301***	.539***	.124
18 Weeks	.810	5.79	.344***	.600***	.819	.009*	5.65	–0.15	.276**	.539***	.156*
16 Weeks	.810	5.80	.336***	.606***	.823	.013**	5.60	–0.20	.257**	.538***	.181**
14 Weeks	.817	5.69	.336***	.612***	.827	.010*	5.53	–0.16	.268**	.548***	.161*
12 Weeks	.819	5.65	.342***	.609***	.831	.011*	5.47	–0.18	.270***	.549***	.164*
10 Weeks	.828	5.51	.327***	.628***	.841	.013**	5.31	–0.21	.253**	.562***	.174**
8 Weeks	.854	5.08	.299***	.670***	.864	.010**	4.90	–0.18	.240***	.604***	.156**
6 Weeks	.861	4.96	.233***	.728***	.874	.014**	4.72	–0.25	.174**	.645***	.178**
4 Weeks	.883	4.56	.204**	.766***	.893	.010**	4.35	–0.21	.156**	.689***	.157**
2 Weeks	.878	4.65	.233***	.740***	.888	.011**	4.44	–0.21	.188**	.658***	.159**

Social Media as Supplement to Survey Data (2)

- Glassdoor
  - Glassdoor is a platform on which employees (current and former) can anonymously post reviews of their employers as well as salary information which users can search; users can apply for jobs through the site
  - Symitsi et al. (2021) improved prediction of household expenditure in the US* by combining survey-based consumer sentiment indices with 5 million employee ratings of their employer's business outlook on the Glassdoor platform

* based on Federal Reserve Bank of St. Louis data

Can Social Media Posts Replace Survey Data?

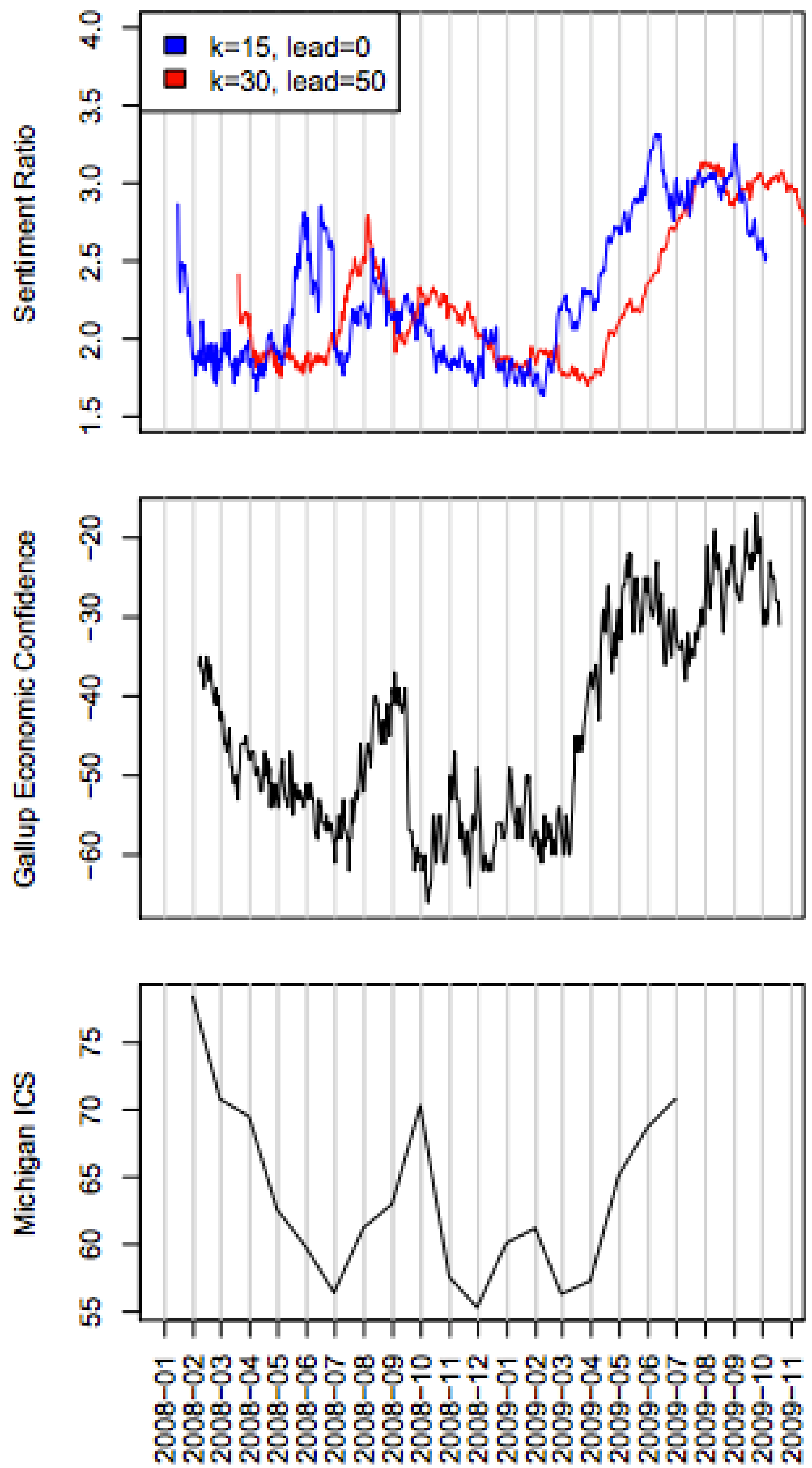
- Prerequisite question is whether the two data sources “tell the same story”
- So far, relevant research has asked if survey responses and social media posts (volume and sentiment) move together <u>over time</u>
- Some success stories replicating survey results with social media data:
  - O’Connor et al. (2010) found that sentiment (positive or negative) of tweets containing “jobs” correlated highly (*r*=.79) with Gallup’s Economic Confidence Index and Michigan’s Index of Consumer Confidence (*r*=.64).
  - Both indices based on answers to responses to > 1 survey questions

O’Connor et al. (2010)

1. Classified each tweet in 2008 - 2009 containing “economy,” “job” and “jobs” as positive or negative then computed sentiment for each day
   a. OpinionFinder assigns sentiment (-1, 0, +1) to each word; a tweet is positive if contains a positive word, negative if contains a negative word
   b. Daily sentiment = (# positive tweets/ # negative tweets)
2. Smoothed over either 15- or 30-day intervals to stabilize data
3. Created lags of -90 to +90 days to account for possibility that one series reflects opinion earlier than the other

O’Connor, et al. (2010):

- Effects clearest for tweets with “jobs”
- With 15-day smoothing and 20-day lead, Gallup-Twitter *r* = .73.
- With 30-day smoothing, and 50-day lead, ICS-Twitter r = .64
- Suggests tweets lead surveys



*survey-based Index of Consumer Confidence

Replacing Survey Data (2)

- However, these patterns have not held up over time and under close scrutiny
- Conrad et al. (2020) replicated O'Connor et al. (2010): *r* = 0.65 for 2008-2009
- But correlations volatile from year to year

Table 5. Correlations Between Index of Consumer Sentiment and Twitter Categories by Year for 2008–2014 Under Settings Used Originally by O'Connor, Balasubramanyan, Routledge, and Smith (2010) for 2008–2009.

Category of Tweet	2008	2009	2010	2011	2012	2013	2014
All tweets	.21	.66	−.03	.54	.02	.28	.41

- And original (2008-2009) finding was fragile, e.g., correlation depended on formula for calculating sentiment of tweets in a day

	$\frac{\textit{Pos tweets}}{\textit{Neg tweets}}$	$\frac{\textit{Pos tweets – Neg tweets}}{\textit{Total tweets}}$	$\frac{\textit{Pos tweets}}{\textit{Pos tweets + Neg tweets}}$
All tweets	0.65	0.00	0.48

Source: Conrad, Gagnon-Bartsch, Ferg, Schober, Pasek & Hou (2020)

When are social media posts more likely to align with survey results?

- The survey responses
  - Sufficient change in survey responses over time for alignment between the series to be possible (Conrad et al., 2022)
- Social media posting
  - Frequency (volume) vs. sentiment, e.g., sentiment makes sense for measuring subjective well-being

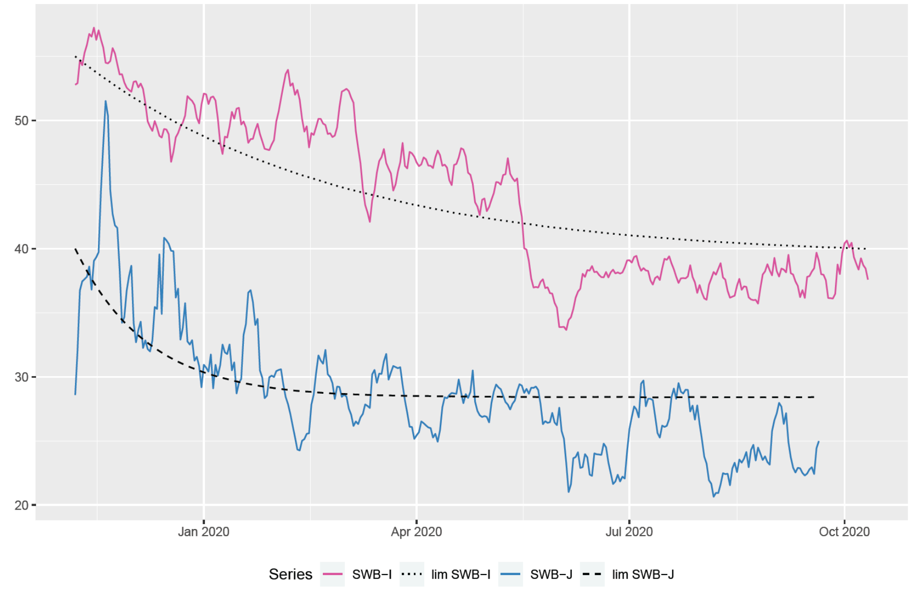


Figure 2: The daily oscillation of SWB-I and SWB-J from November 2019 to 10 October 2020 for Italy and 20 September for Japan, and their trend estimates ("lim SWB-I" and "lim SWB-J"). The vertical axis represents the value of the index on the [0-100]% scale. The total volume of tweets is 13,975,242 for Italy and 12,907,902 for Japan.

- But less so for a question about likelihood of a behavior, e.g., "Have you heard of the 'American Rescue Plan'?" for which sentiment of tweets might be positive or negative, irrespective of likelihood of having heard of the plan

Source: Iacus & Porro (2022)

Summary: Social Media for Social Measurement

- Social media posts are not the same as survey responses
- Promising for problems which are hard to address with traditional survey methods
- Promising initial results from supplementing survey data with social media data
- Jury is still out with respect to using social media content in place of survey data
- We’re still getting started